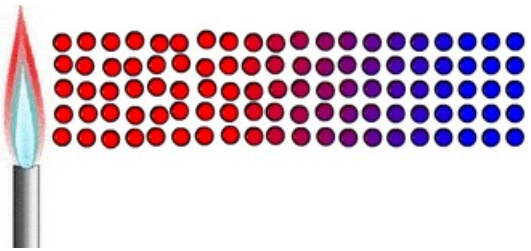
Knowledge Organiser
Year 8: Heating and Cooling – Transferring Heat

Temperature vs. Heat:
- Temperature is a measure of how hot something is
- Heat is a measure of the thermal energy in an object
- **Temperature** is measured in ºC
- **Heat energy** is measured in J

How does heat energy behave?
- Heat energy will flow from hotter to colder regions
- It can flow by conduction, convection or radiation

How does conduction work?
- Conduction is the movement of heat by vibrating particles
- The particles do no move from their position, they vibrate around a fixed point
- When part of a solid absorbs heat energy the particles vibrate faster
- The vibrations are passed between particles

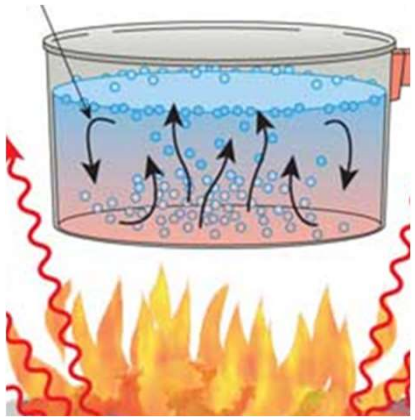
Diagram which shows the **transfer of heat energy by vibrating particles**



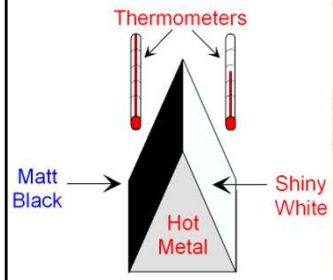
Which are the best conductors?
- Solids are the best conductors of heat
- Their particles are close together so can easily pass on vibration
- Metals are even better because they have **free electrons** which can move throughout the substance carrying thermal energy with them

How does convection work?
- Convection is the transfer of heat **by moving particles**
- When particles in a fluid they gain energy and **move away** from each other
- The particles are then **replaced** by ones with less energy

Diagram which shows the **transfer of heat energy by moving particles** in a fluid



How does thermal radiation work?
- Infrared (thermal) **radiation** is a type of **electromagnetic** radiation
- When infrared radiation is absorbed by an object it is heated and its temperature rises
- Unlike conduction and convection it does not require particles so can work in the **vacuum** of space

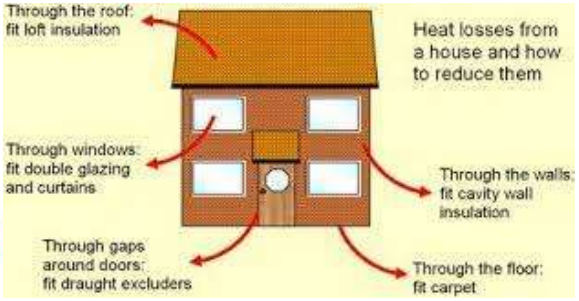


What types of surfaces absorb thermal radiation the best?
- **Emit** means to give out energy
- **Absorb** means to take in energy
- Dark surfaces are better at both absorbing and emitting heat energy than light ones
- **Matt** (rough) surfaces are better at absorbing and emitting heat energy than shiny ones
- A black car would both **heat up** more quickly on a hot day and **cool down** more quickly on a cold day

How does heat escape our homes?
- It is transferred by conduction through the walls, floor, roof and windows
- It is also transferred by convection as cold air can enter through gaps in doors and windows

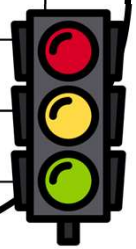
How do insulation methods work?
- **Double glazing** prevents heat loss by conduction as the air between the panels is a poor conductor
- **Cavity wall insulation** inside walls is full of air pockets, which also prevents conduction
- **Carpets, curtains** and **draught excluders** prevent the flow of cold air through convection

Diagram which shows ways to prevent loss of heat energy from the home



RAG Yourself!

Learning Outcomes;	RAG
Describe how conduction transfers thermal energy	
Define insulator and conductor with examples	
Describe how convection transfers thermal energy	
Describe how surfaces affect thermal radiation	
State methods used to reduce heat transfer at home	
Explain insulation methods in terms of reducing conduction, convection or thermal radiation	



Knowledge Organiser
Year 8: Heating and Cooling – Transferring Heat

Temperature vs. Heat:

How does heat energy behave?

How does conduction work?

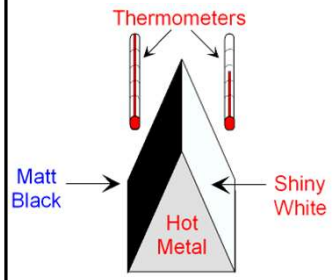
Diagram which shows the **transfer of heat energy by vibrating particles**

Which are the best conductors?

How does convection work?

Diagram which shows the **transfer of heat energy by moving particles** in a fluid

How does thermal radiation work?



What types of surfaces absorb thermal radiation the best?

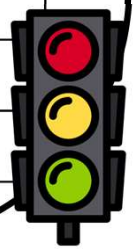
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